

Sukhsimran Singh

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Jalandhar, Punjab, India

EDUCATION

- I.K. Gujral Punjab Technical University** Kapurthala, India
 - B.Tech in Computer Science and Engineering; CGPA: 9.01 / 10* 2023 – 2027

TECHNICAL SKILLS

- Languages:** Python, SQL, C++
- Generative AI & LLMs:** LangChain, LangGraph, RAG, Hybrid RAG, Prompt Engineering, HuggingFace, Embeddings, Groq, ChromaDB, FAISS, MCP Servers, Agentic Workflows, AI Agents
- Machine Learning & AI:** Scikit-learn, TensorFlow, XGBoost, Neural Networks, Deep Learning, Transformers, Feature Engineering, Model Evaluation
- MLOps & Deployment:** FastAPI, REST APIs, Model Deployment, PostgreSQL, SQLite, Git, Docker, Data Pipelines, Model Serving, Render, Vercel
- Data Science:** Pandas, NumPy, Matplotlib, SHAP, EDA, NLP, Time Series Analysis, Tableau, Data Analysis

EXPERIENCE

- FlyRank AI** Jul 2026 – Present
 - Backend AI Engineer Intern*
 - LLM & Agentic Systems:** Built LLM-powered agentic workflows integrating Retrieval-Augmented Generation (RAG) pipelines with PostgreSQL and REST APIs for production environments.
 - AI Backend Development:** Developed scalable backend services and production-ready REST APIs in Python and FastAPI for AI-powered applications.
- Internshala Training (IITM Pravartak + NSDC)** Jun 2025 – Jul 2025
 - Data Science Trainee*
 - Machine Learning Pipelines:** Built end-to-end machine learning pipelines (data preprocessing, exploratory data analysis, feature engineering, model training, and evaluation) on real-world datasets.
 - Recognition:** Recognized as **Top Performer** among trainees for delivering high-quality machine learning solutions and demonstrating strong analytical and problem-solving skills.

PROJECTS

- PranRakshak AI: Intelligent Clinical Decision Support System** Live | [GitHub](#)
 - FastAPI, React, PostgreSQL, LangGraph, RAG, Groq, SHAP, OCR, Render, Vercel*
 - Clinical AI System:** Engineered an early sepsis detection model on the PhysioNet CinC 2019 ICU dataset (**40K+ patient records**) using Sepsis-3 guidelines, SHAP-based explainability for clinical risk prioritization.
 - Generative AI & Deployment:** Built a RAG-powered clinical assistant with OCR-based medical report understanding using Groq LLMs, and deployed the full-stack application (FastAPI, PostgreSQL, React) on Render and Vercel.
- DisasterSense AI: AI-Powered Disaster Intelligence Platform** Live | [GitHub](#)
 - FastAPI, React, PostgreSQL, FAISS, Sentence Transformers, Google Gemini*
 - AI Disaster Platform:** Built a full-stack disaster intelligence platform (FastAPI, React, PostgreSQL) for emergency assistance, incident reporting, and AI-powered disaster image analysis.
 - RAG & Vision AI:** Implemented FAISS-based Retrieval-Augmented Generation with Sentence Transformer embeddings and Gemini 2.5 Flash, integrating Gemini Vision for automated disaster severity assessment and emergency guidance.
- OmniRAG AI: Multi-Document Research Assistant** Live | [GitHub](#)
 - FastAPI, Next.js, LangChain, ChromaDB, HuggingFace Embeddings, Llama 3*
 - RAG Architecture:** Developed a multi-document Retrieval-Augmented Generation system with isolated retrieval pipelines (LangChain, ChromaDB, HuggingFace embeddings, Llama 3), supporting documents for contextual AI conversations.
 - Scalable Deployment:** Designed a scalable document ingestion and semantic retrieval workflow (FastAPI backend, Next.js frontend) for real-time document management.
- Resume Screening AI: NLP-Based Candidate Evaluation System** Live | [GitHub](#)
 - FastAPI, Sentence Transformers, BERT, Scikit-learn, Streamlit*
 - AI-Powered Resume Evaluation:** Built and deployed an NLP-based resume screening system using BERT and Sentence Transformer embeddings to semantically match resumes with job descriptions, providing section-wise candidate scoring and skill-gap detection via a FastAPI REST API.

HONORS AND ACHIEVEMENTS

- Top 15 Finalist, Innovate with TRAE – NIT Jalandhar
- Finalist, Hack On Hills – NIT Hamirpur
- Participant, Hackathon 4.0 – GNA University

CERTIFICATIONS

- Data Science:** IITM Pravartak & NSDC – [Certificate](#)
- Data Science, Machine Learning & NLP:** Udemy – [Certificate](#)
- Generative AI with LangChain & HuggingFace:** Udemy – [Certificate](#)
- LangChain:** [Certificate](#)

LEADERSHIP

- Core Team Member,** Upstarts – Technology & Startup Club, IKGPTU 2024 – Present
- AI/ML Hackathon Participant,** National-Level Innovation Competitions 2024 – Present